37

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After Zero, New Numbers

DPL-01

A Day Plotted Logarithmically

After Zero, New Numbers

Rest Area, Laplace Museum

Visitor I: Alright now, let me get a good look at this thing. So, the special exhibition area is located in Zone C. It looks like we need to take this path to get there.

Visitor II: I just can't believe I'm finally getting to see the stone with my own two eyes!

Visitor III: Yeah, I remember those critics on TV calling it all a hoax. Let's see what they have to say for themselves now.

The Laplace Museum, a place rich in history and heritage. For a long time, it has been silent, but now, it buzzes once more with shuffling feet and whispered curiosity.

Thirty-seven: So, this is the promotional poster for the special exhibition.

Thirty-seven: I like the simple geometric lines. Their arrangement is just right.

Three young people sit at a round table in the museum's refreshments area. One of the girls, dressed in peculiar, flowing robes, draws curious glances from passersby.

However, her eyes do not budge—they are fixed on the poster in her hand. Its title reads: "LAPLACE MUSEUM—SPECIAL EXHIBITION: THE MIRACLE STONE FROM 4.5 BILLION KILOMETERS ACROSS THE COSMOS."

A small area circled at the bottom reads: Special "Star of Wonder" themed ice cream—available during exhibition only.

Shopkeeper: Sorry for the wait, folks! Here's your ice cream!

Thirty-seven: I don't think we waited too long, but thank you all the same.

Shopkeeper: Come on, now! You and your friends saved my shop! I owe you big time!

The two other young people at the table also receive their ice cream.

Ezra: Looks yummy—even better than the poster.

Mesmer Jr.: I wouldn't usually eat this kind of thing. All the same, though, I never thought things would turn out the way they did.

Ezra: Miss Miller told me she considers the survival of her ice cream shop to be the result of your hard work, but I still don't know the whole story.

Ezra: And then, there's Miss 37.

Ezra: Everyone at the computing center has been talking about you. They say you're a brilliant mathematician! I'm honored to finally be meeting you in person.

Thirty-seven: Please, there's no need for formalities. You can call me 37, and I'll call you Ezra, if that's alright? I've heard a lot about you, too.

Thirty-seven: You're precisely as I thought you'd be. Your real and imaginary parts are neat numbers —not that I dislike that at all. I've never had an imaginary number for a friend before!

Ezra: Uh, imaginary numbers? Like the ones Descartes proposed in the 17th century?

Mesmer Jr.: You'll get used to it. I had no clue what she was saying when I first met her. To be honest, half the time, I still don't.

Ezra: Haha! I'm sure that must be an interesting story. I still have some time before my lecture. I'd love to hear it, if you'd like.

Thirty-seven: Certainly. I think we should start with my theorem that our heads in tangent with the table.

Ezra: Sorry?

Mesmer Jr.: I warned you.

Mesmer Jr. sighs.

She'd rather just finish her ice cream in peace, but it seems the task of telling the story sensibly has been forced upon her.

Mesmer Jr.: Well, it all started with an old classmate—Vertin.

Mesmer Jr.: It was all because of her that I got mixed up with 37, and a series of peculiar events followed.

Mesmer Jr. begins her tale, and time is rewound to three days ago.

This is a story about a museum, ice cream, distant stars, and numbers.

Lobby, Laplace

Mesmer Jr.: *sigh* What was she thinking? Do I look like a babysitter?

Mesmer Jr.: Who even uses this kind of thing nowadays?

The therapist looks down at the scroll in her hand, like something out of Ancient Greece.

Mesmer Jr.: It's none of my business.

Mesmer Jr.: Excuse me, I'm looking for someone here from the Foundation. She came to headquarters for advanced training.

Laplace Researcher: Incredible! So, that was the right value ... eh? Huh? What was that?

This particular self-absorbed target for her inquiries is not the best choice, but then finding a "normal" researcher at Laplace is never a simple task.

With the lobby uncommonly empty today, it seems there is no better option available.

Mesmer Jr.: I said I'm looking for someone here from the Foundation. She came to headquarters for advanced training.

Laplace Researcher: Do I look like a receptionist? Go ask someone else! I don't have time for—hang on.

Laplace Researcher: Ugh, you're looking for 37, too?

In both number and name, it seems they're both looking for the same figure, and that they're not alone.

Mesmer Jr.: Too? Yes, I'm looking for a girl named 37.

The researcher's eyes widen, a fear response.

Laplace Researcher: No. No, no, no.

He begins moving toward the hallway as fast as his uncoordinated, noodle-like limbs will carry him.

Laplace Researcher: I was here first! You! You get behind me. Wait your turn!

He runs off. As the echoing of his footsteps fades from the lobby, the lone therapist is left standing there, bewildered.

Mesmer Jr.: What?

Hallway, Laplace

Mesmer Jr.: What's going on here?

Researchers are spread across the corridor like an outbreak of gray-coated mold.

Some are locked in heated discussion, while others sprawl out on the floor or lean against the walls, writing, notating, and sketching out figures and graphs. All of them enthralled in the kind of manic frenzy Mesmer Jr. recognizes but certainly does not condone.

Mesmer Jr.: So, that's why the lobby's so empty today. Emptier than usual, anyways.

She pauses there a while, as a constant stream of researchers clutching their papers rush into the room at the end of the hall, or go running off out of it, waving their seemingly solved problems.

Laplace Researcher: Yes! She's got it! That girl really is a genius!

Mesmer Jr.: This place never changes. Egghead scientists losing their minds at the slightest provocation.

Fortunately, the researchers, glued as they are to their place in line, take no notice of her.

She takes a deep breath and heads to the room at the end of the hallway.

Research Lab, Laplace

Piles of paper and scattered geometric models almost bury the entire desk, and the small office is crammed with researchers, packed like jittering cans of liquefied anxiety.

They wave their papers in the air, surrounding the desk like hungry Glawackus.

Laplace Researcher I: Hey, 37, look at this! I calculated it using the method you suggested last time. Here's the new data. But now a new problem's cropped up.

Laplace Researcher II: Wait! I'm not done yet! Wait your turn!

Laplace Researcher I: I've been waiting all day, you buffoon! Swallow your pride, and just accept 37's algorithm is simpler than yours!

Laplace Researcher II: What? Tosh! Tosh, drivel, and baloney! Just look! Look closely, you—

Mesmer Jr.: What kind of mess have I walked into now? This place is a battlefield in egghead hell.

The therapist draws a sharp breath, pulling her head out of the chaotic abyss.

X: Personally, I'd be more inclined to employ the metaphor of "valiant heroes catapulting themselves into a courageous confrontation to earn the favor of their goddess."

Mesmer Jr.: X.

After the initial surprise of hearing the familiar voice, Mesmer resumes her usual indifferent expression.

Mesmer Jr.: Do you intend on becoming one of these "heroes"?

X: Rather than genuflecting to win the favor bestowed by this "goddess," I intend only to extend an invitation her way, to join me in spectating a sumo-wrestling match.

X: However, encountering a friend here—that's you, Mesmer—I must admit, I have indeed been taken quite by surprise.

X: I didn't take you to be the type to enjoy these gory spectacles; even when faced with challenges, you lean toward resolving them independently. Oh.

The papyrus scroll cradled in the brunette's arms is an obvious clue.

The young researcher may excel at constructing complex systems, but he is just as adept at circumventing obstacles and distractions to find the truth.

X: Was this the Timekeeper's doing?

Mesmer Jr.: She certainly knows how to use people to get what she wants.

Not long ago, while Mesmer Jr. was on official business at the Foundation, she encountered her now illustrious former classmate.

At that time, she was hurrying down the hall, on the phone with some unknown party. She always seems so busy.

Vertin: Yes, I'm on my way. I'll be there soon, Mesmer Jr.

Mesmer Jr.: Vertin.

Vertin: I must say, I'm actually glad to see you. I have a favor to ask of you.

The busy star of the Foundation makes her request—to deliver an important letter on her behalf to a colleague training at the computing center.

Mesmer Jr.: They have this thing these days called Critter Express.

Vertin: I'm afraid Carbuncles won't be able to confirm what's going on with 37 for me.

Vertin: She's said that she's fine, but she's never left her island before, so I worry that her "fine" may not really count as "fine."

Vertin: I was hoping you might help confirm just how "fine" she is now—as the most trustworthy person I know at the computing center.

Mesmer Jr.: ...

How amusing. She seems to have forgotten what transpired in this very hallway, but Mesmer Jr. knows that is impossible.

Is this a trap? Psychological manipulation? A true gesture of goodwill? Or perhaps ...

She should have said "no."

Mesmer Jr.: I really don't know why she insisted on asking me.

X: I have always trusted her judgment, particularly when it concerns people.

Mesmer Jr. doesn't wish to dwell on the topic. Her eyes reach for the room down the hallway.

Mesmer Jr.: Anyway, how would you describe things around here? Vertin leaves this girl with you for a second, and now it looks like she's being eaten alive. How long has this been going on?

X: Since the day she arrived at the computing center. Do you recall the problem Baird left on the blackboard to the left of the lobby over a year ago? The one no one could solve?

X: That girl passed by during her first-day perambulations, and two seconds later, bam! She had the answer all chalked out!

X: Since then, this hallway has become the rowdiest strip in the entire LSCC.

Mesmer Jr. looks at the researchers coming and going, a frown growing more evident on her face.

Mesmer Jr.: There are researchers from every department here. Are you telling me that this one girl has insights into every field of research compelling enough to cause all this?

Mesmer Jr.: It's impossible.

X: Oh, no, no, no. That's not it. She's barely aware of most of our research. All she deals with is "mathematics."

Mesmer Jr.: Mathematics.

She repeats the word.

Mesmer Jr.: Of course. I've heard that "that island" was a hotbed of math geniuses.

X: Exactly. Just as dear old Leonardo pointed out long ago, mathematics is the foundation of all sciences, but most people have failed to grasp this universal truth.

X: But this girl is not like most people. She doesn't know anything about chemistry, astronomy, or physics.

X: But she can solve any single one of their problems if she can reduce it to a mathematical problem.

X: I'm not overstating her abilities here. There's yet to be a math puzzle she's encountered that's eluded her computational comprehension.

Mesmer Jr.: Really? She does all that with just mathematics?

X: I, too, harbored reservations until I observed it in action. Maybe I could offer an example for your elucidation.

X: The original progenitors of "mathematics" employed a fascinating sexagesimal numeric system, yet this is not the most extraordinary aspect of their knowledge.

X: The most remarkable aspect was that their frameworks lacked both the notion of zero and of decimals, yet they still managed to produce accurate results.

X: The pinnacle of our most advanced arcanum pales in comparison to the skills demonstrated by that remarkable girl.

Mesmer Jr.: Exaggerating as always, I see. We both know how they did it.

Mesmer may not enjoy drawing attention to her intellect the way her peers do, but there is no doubt that she belongs among them.

Mesmer Jr.: 1, 10, 100, and 0.1, all represent 1 when written in their method. Whenever they needed to use this number in their calculations—like if they wanted to multiply it by 100—in theory, they'd end up stuck, with no way to proceed.

Mesmer Jr.: Since they used cuneiform symbols but not the actual numbers they represent, they had no way of knowing what numbers the symbols stood for.

Mesmer Jr.: But they didn't need to know.

Mesmer Jr.: In their abstract mathematical world, whether the answer was 100, 1,000, or 10,000 ... ultimately, by following the same rules, they'd arrive at the same answer—"1."

The therapist shakes her head.

Mesmer Jr.: There's nothing magic about it.

Mesmer Jr.: Please step aside. Sorry, today's Q&A session is over.

Laplace Researcher: By what right do you think you've got a monopoly on 37's time? This is a travesty! All of Laplace will suffer, you hear?! You're holding back global scientific progress.

She slams the door. Global scientific progress will have to wait.

Mesmer Jr.: Alright. Now ...

Mesmer Jr. clears a path to the desk.

Beneath the desk, next to a chair rendered an impromptu bookcase due to the pile of papers stacked on it, hides a head of tangled light-colored hair.

Thirty-seven: So, it can be proven that when n is greater than 2, there are no positive integer solutions ...

Thirty-seven: This proof is correct! But his method is different from ours. Incredible!

Platonic Solids

DPL-02

Thirty-seven: Yes, this proof is flawless, and even better. It's correct!

Thirty-seven: Why didn't anyone on the island realize there was another way? I need to note this down. 6 will definitely want to know about this.

Mesmer Jr.: Hello, 37. Sorry to interrupt.

Thirty-seven: How regrettable. This hypothesis was so intriguing, yet ultimately untrue. It's always a heartbreak to disprove an elegant theory.

Thirty-seven: The conjecture regarding planets and Platonic solids was similar. It was so perfect. Everyone wished it might be true, but it wasn't.

Mesmer Jr.: ...

The girl beneath tangled hair is absorbed in her own world.

Perhaps it is just her imagination, but there is a trace of melancholy in the girl's voice. But who could feel sad over a mathematical proof?

Thirty-seven: There must be some perfect order, a function through which all things can be predicted. We just haven't found it yet.

Thirty-seven: Error, then, doesn't really mean failure. Within each failure may lie the key to the Truth.

Thirty-seven: Yes, that must be it, ow!

A dull thud. An expected result.

X: That is why children are usually not permitted to play underneath tables. Perhaps you should consider utilizing your chair for its intended purpose, 37.

Mesmer Jr.: I probably should've asked before, but why exactly did you follow me in here?

X: Oh, don't be like that. I didn't say anything about carelessly abandoning my previous intention of extending an invitation to our computational celebrity to partake in spectating a muscle-bound sparring competition.

When it comes to X, the more long and nonsensical words he puts in a sentence, the more likely he's trying to hide something.

But prying the truth from him would be effort she'd rather not waste. She's better off ignoring him.

He's one small spore of gray mold left in the room, that's all.

Mesmer Jr.: Are you okay?

Thirty-seven: Ugh, I'm fine.

Thirty-seven: If we consider my head to be a circle and the edge of the table as a straight line, then no matter where I actually hit it, at the moment of collision, my head was only briefly tangential to the table.

Thirty-seven: Not intersecting, not separate, but in tangent. This is a very good positional relationship. From the center of the circle—my head's ideal model center—a line through the tangent point must be perpendicular to the table.

Thirty-seven: The sine of 90° is 1, the cosine is 0, and the tangent is infinity! There's no better angle than 90°.

Thirty-seven: So, I'm totally fine. It doesn't hurt at all.

Mesmer Jr.: Fine.

In truth, her peculiar explanation, coupled with the tears welling in her eyes, is not all that convincing.

Thirty-seven: Hmm? You're a new face.

It's an odd way to get across the question. Most might say "who are you" or some other politesse, but then this girl is not "most people."

Thirty-seven: What numbers have you brought me? Let's have a look at your problem.

She extends her hand to the therapist. It appears this is the protocol she has established with the Glawackus from earlier.

Mesmer Jr.: The number I've brought is "6."

She places the papyrus letter in the girl's palm.

Thirty-seven: 6 ...?

Thirty-seven: Yes. Hmm, 6 is a good number. A familiar number.

Thirty-seven: Hmm, what kind of proof do you wish to make with this number?

The girl's mathematical babbling slows as she gazes deeply at the papyrus scroll in her hand, lost in contemplation.

Mesmer Jr.: I don't have a problem.

Mesmer Jr.: This is a letter from someone named "6." Vertin asked me to give it to you.

Thirty-seven: Vertin? And 6!

Her eyes light up. She grabs the outstretched scroll and unties it in a flurry of movement.

Thirty-seven: Let's see what 6 wrote. Hmm, first question: Are you getting used to life over there?

Thirty-seven: I am indeed. Meals are delivered to me daily through an intriguing pneumatic tube system, and they don't serve me any beans. This allows me to focus on my research without any distractions.

Thirty-seven: Second question: Have you gotten sick? No, I'm in good health. Though, without Sophia to comb my hair, it has gotten a bit matted. But that's alright. I just cut out the parts I can't untangle.

Thirty-seven: Have I learned anything new? Yes, I've learned so many new things and built up a collection of newspaper clippings I plan on sending back to the island.

Thirty-seven: My research? Well, I've gathered a lot of numbers, and solved many problems, and ...

Thirty-seven: And ... hmm.

The girl stares at the letter in her hand, falling silent.

Mesmer Jr.: Is something wrong?

Thirty-seven: To be honest, I don't think my research is going well at all.

Thirty-seven: My hypothesis was that if I ventured to the world beyond the island, I might encounter many more numbers than before.

Thirty-seven: I theorized that so long as I obtained enough numbers, no matter how complex the function, I could graph it and calculate its formula, or at least obtain the clues necessary to solve them.

Thirty-seven: Numbers are the keys to the Truth. I must decipher them.

The girl says the word with absolute conviction.

Mesmer Jr.: The "Truth"? Is that the topic of your research?

Thirty-seven: Yes.

The response comes without the slightest ambiguity.

Thirty-seven: Last time, I failed. I got the answer, but I was too slow, and it wasn't enough. It wasn't complete. And ... bad things happened.

Thirty-seven: That's why I need to find a better answer.

Thirty-seven: A more complete Truth, then the bad things won't happen again, and everything will function as it should, in the most ordered manner.

Mesmer Jr.: Is that so?

She says it all with such sincerity that, as bizarrely grandiose as it sounds, Mesmer knows the girl is completely serious.

But how could she even begin to evaluate such an overly ambitious goal?

Before she can respond, the girl lowers her head.

Thirty-seven: I've run into a problem.

Thirty-seven: I've encountered many more numbers, but I still can't decipher the functions at work behind them.

Thirty-seven: It feels like ... Oh, are you familiar with "hopscotch"?

Mesmer Jr.: The kids' game?

Thirty-seven: Yes. Sophia taught me how to play it. I think my situation is a bit like hopscotch.

The name is totally unfamiliar to her, but the girl seems unaware even of the need to explain that much.

She lowers her gaze and takes a step forward on the laboratory floor.

Thirty-seven: Heh!

Thirty-seven: See? Numbers can be found within all things. So now, I'm currently in this square.

Mesmer Jr.: Sure ...

As far as she can see, there are no squares on the floor. And what does she mean by "numbers can be found within all things"?

Thirty-seven: And behind the numbers lies the path to the Truth. But ...

The girl looks at the floor in front of her, shaking her head.

Thirty-seven: I can't see where the next square is drawn, so I'm stuck here, unable to move forward.

Mesmer Jr.: I understand.

She doesn't understand at all. A part of her wishes she could say, "Well, keep trying! I'll be leaving now."

But as she watches the girl, who was so exhilarated reading that letter and now appears so despondent, she cannot bring herself to say it.

X: You see, that's the "why" of it.

The therapist's glance signals for the mold behind her to remain silent.

Mesmer Jr.: I'm not sure of the specifics, but it looks like you've hit a block.

Mesmer Jr.: This is very normal. I'm sure nearly everyone in the computing center has been in your shoes before.

Thirty-seven: Do you mean it? Everyone here has been through the same thing? This is the first time I've felt so hopeless. Up until now, even if I didn't have the answer right away, I never felt upset about it.

Thirty-seven: I don't really understand it.

Mesmer Jr.: ...

The experienced artificial somnambulism therapist needs only a glance around the room to identify the problem.

Closed windows, dusty corners, disorganized files, a blanket thrown on the sofa, and stainless-steel pneumatic food delivery cans scattered all over the place ...

Mesmer Jr.: When was the last time you went outside?

Thirty-seven: Uhm, that's strange. I can only provide an approximate value. Nothing precise.

Mesmer Jr.: I thought as much. Listen, maybe you could go out with your frie—

Mesmer Jr.: ...

Mesmer Jr. holds back the second half of her sentence. She knows something of the girl's background; she left her home and came to the computing center alone to carry out her research.

With everyone from Team Timekeeper preoccupied with other tasks, 37 most likely doesn't have any friends to "go out with."

Thirty-seven: Perhaps I could?

Mesmer Jr.: Perhaps you could go out for a walk or something.

Her professional ethics demand that she offer this non-committal, all-purpose suggestion.

Thirty-seven: A walk?

The girl thinks for a moment, then nods.

Thirty-seven: You make a good point. A walk might help me to progress my research. It is a solid hypothesis, at least.

Thirty-seven: Previously, when we needed inspiration, we might take a walk on the beach or calculate the trajectory of seagulls while relaxing in the sun.

The suggestion is accepted, but the girl still appears hesitant.

Thirty-seven: Perhaps I shouldn't let such trivial worldly concerns hold me back, but I've discovered that there are no beaches nearby, and no seagulls either.

Thirty-seven: So that's why I'm not all that enthusiastic about going out for a walk.

Mesmer Jr.: ...

Some people are blind to their own emotions. They disregard their own feelings and immerse themselves in solving other people's problems.

This is true of one of her classmates and of the girl standing before her. Mesmer Jr. doesn't know much about 37, but she surmises the problem is not just an absence of sand and seagulls.

Her work was over. She shouldn't even be here anymore.

Mesmer Jr.: Maybe you could go somewhere else?

Thirty-seven: Do you have any suggestions?

Mesmer Jr.: Well, you could try ...

A blind spot emerges. Mesmer is not quite skilled at "relaxing" either.

But by luck, the mold still growing silently in the room happens to be quite an expert in the field.

X: Ladies, it appears you might require some advice. Would you like to hear my two cents?

Deviation

DPL-03

Deviation

X: And now, lucky me, I'm honored to serve as the escort for you two ladies.

Mesmer Jr.: You're late.

X: Sorry, but at least I'm not the last one we're waiting on.

The fourth chair at the round table is pulled out. The young researcher seats himself with his characteristic smile.

X: Sounds like you're expounding on our recent narrative. Ezra, my friend, can you deduce our current destination?

Thirty-seven: The question is simple, but as usual, we'll need a hint: we are currently within the domain of this function.

Ezra: Haha, I think I'm beginning to understand Miss 37's—I mean, 37's sense of humor.

Ezra: It's here, isn't it? The Laplace Museum.

Thirty-seven: Correct!

Gallery, Laplace Museum

Thirty-seven: I've heard of "museums" before.

Thirty-seven: This place is like a number line marked with important points. We can't mark all the good numbers because they're infinite.

Thirty-seven: But we can still pick some representative ones, so when we look at this number line, we can see their progress and relations. It's a good approach.

Thirty-seven: What's in this glass case? Is it some kind of metal ore?

X: That's lepidolite. It contains metal, but we typically classify it as a rare earth mineral. People often confuse it with zircon, but I must assert there is a substantial difference.

X: But I thought you'd be more intrigued by the azurite adjacent to it. See that ethereal blue color. It typically captures the eye before its neighbor.

Thirty-seven: All the crystals inside have three axes perpendicular to each other, but the lengths are not the same. It's not a perfect shape.

Thirty-seven: This one is different. Look closely. If we took a cross-section, it'd form a perfect square. The four faces of these crystals are symmetrical ... Wait!

Thirty-seven: Look at this, the crystals. There's no mistaking it. They're made of dodecahedrons, and regular dodecahedrons at that! This mineral has the most perfect shape in this display case!

X: Then allow me to thank you on behalf of this garnet. In fact, I also have a fondness for this stone.

X: An endearing appearance paired with peerless practicality. Have I mentioned I possess many small spheres of this mineral, of varying sizes and cuts?

X: I often select one based on my mood. Observing those exquisite little objects roll on the tracks is a real delight.

Thirty-seven: You mean, you have lots of perfect spheres? May I see your collection?

X: Absolutely. I'm glad we share a similar interest in them.

The smiling researcher looks at the young mathematician, pondering over her for a moment.

X: You've got a unique perspective on the exhibits here, but I think there are some others you'll find even more fascinating.

X: Are you interested in the insect exhibition?

Thirty-seven: Do they have any weevils? I've heard that they contain a perfect representation of the golden ratio!

The girl bounces toward the insect hall like a mint-green balloon.

Mesmer Jr. doesn't take part in their animated discussion about minerals, bugs, and geometry. She remains, as ever, a model visitor, quietly appreciating the exhibits and offering only occasional whispered thoughts.

Mesmer Jr.: This place has changed a lot since the last time I was here. I don't remember any of these exhibits.

X: After our mutual friend Ezra took over the mantle of curator, he has exerted considerable effort into this place's development.

X: He reconfigured the layout of the halls, increased the number of exhibits by nearly half, and greatly expanded the range of the items on display.

X: Check out this golem dung, for example. Before Ezra, you'd never see something like this here, and at most, the exhibit's description would merely state that "over 90% of rock golem dung is composed of rock."

X: Not to disparage, but the old curators were the type to say, "Oh, no, we can't have this sort of thing here. It's entirely inappropriate. Get that out of my display!"

X: But here it is: rocky golem dung, displayed in the world's finest museum.

Scanning over the new exhibits, Mesmer finds herself agreeing with her colleague for once.

Mesmer Jr.: I heard the Laplace Museum's workload has doubled since the "Storm" began. So much for a cushy, prestigious job. Now, it's a real challenge.

Mesmer Jr.: Each time the "Storm" approaches, the museum's exhibits must all be moved to a safer location. Then, once the "Storm" passes, there's the tedious task of taking inventory.

X: Everything devolved into chaos after the "Storm."

X: Painstakingly achieved results obliterated overnight ... After living through it several times, it's difficult not to lose hope.

Mesmer Jr.: But this place has been restored to life.

Mesmer Jr.: Ezra is passionate, inclusive, and pragmatic. Appointing him as curator was the right deci —

A loud explosion interrupts her praise.

Thirty-seven: Huh?

Mesmer Jr.: What was that noise? What's going on here?

Office Hallway, Laplace Museum

In the office area at the rear of the museum, thick smoke billows from a room around the corner.

Thirty-seven: Was that explosion part of an exhibit? I don't like it!

Mesmer Jr.: No. I'm certain it wasn't.

Amidst the confusion, a familiar figure emerges from the smoke, and Mesmer Jr. instantly sees red.

Medicine Pocket: Damn it! I knew the data was way off! Those dumbasses must have volcano ash for brains!

Medicine Pocket: Fortunately, I came prepared. Try this on for size!

The instigator mutters while pulling out a terrapin bulb, roughly stripping its leaves before tossing it into the smoky room.

In a matter of moments, the room ceases to spew smoke into the air, much to everyone's relief.

Medicine Pocket: I'm so done. Couldn't that heat-resistant layer have lasted just a couple more seconds?

Mesmer Jr.: You're not done with me, you aren't! How about you explain why you're here and just what you think you're doing?

Medicine Pocket: Isn't it obvious? I'm making cupcakes, of course.

X: Cupcakes? I do love cupcakes, especially with a cup of fresh-brewed coffee.

Mesmer Jr.: Stop joking around, Medicine Pocket. What were you thinking, bringing your insane experiments here?!

Mesmer Jr.: Does Ezra know about this? How in the hell—I just can't believe you!

Mesmer Jr. has no patience for Medicine Pocket's less than reasonable excuses today.

She feels the need to retract her praise for the new Laplace Museum curator. It's clear to see that the young man's overly generous attitude has opened the door to certain shameless individuals.

Mesmer Jr.: Fine, you know what? Do whatever you want.

The therapist makes her decision. A wise, rational, responsible decision.

Mesmer Jr.: Let's keep moving, 37. Just ignore this person, and this unpleasant little episode, and continue our tour of the museum.

Mesmer Jr.: Of course, if the "cupcake lover" wants to stay behind, he's more than welcome to.

Mesmer Jr.: After all, sometimes you need to be careful about the company you keep. Right, 37?

The mint-green balloon she reaches for is gone, having floated over to the beagle.

Thirty-seven: You mentioned the data being off. What data? May I examine it?

A variety of horrible scenarios flash through Mesmer's imagination. Of course, anything to do with "numbers" seems to carry a special allure to her companion.

Medicine Pocket: First of all, who the hell are you? And second, what's it to you—wait.

They sniff the unfamiliar balloon up and down with curiosity.

Medicine Pocket: Oh, you're the one with numbers for a name. I've heard of you.

Thirty-seven: That's totally inaccurate. There are infinite sets of numbers in the world, but I'm 37—just one number.

Medicine Pocket: Uh, fine, whatever, just show me what you can do, Number Brain! Come on over here, and take a look at this for me!

Thirty-seven: Wow!

The mint-green balloon is whisked away by the beagle. The horror begins to feel real.

Mesmer Jr.: I think it's about time you answered my question: What exactly are you researching?

The therapist's question is directed at the late-arriving last participant of this ice cream roundtable.

But the individual being questioned merely pulls out a chair, sits down, and starts devouring their ice cream with gusto, clearly unfazed by the accusation.

Medicine Pocket: Why don't you exercise that big brain of yours and take a guess? If you can't figure it out, well, I guess you're gonna have to settle with the "cupcakes" explanation.

Thirty-seven: I looked at Medicine Pocket's data; it seemed to be concerned with some kind of plant or maybe an insect? Sorry, I was only paying attention to the numbers.

Ezra: Medicine Pocket is trying to find a way to artificially cultivate Ophiocordyceps Icarus, a precious herb that only grows in extreme volcanic environments, with very limited annual yields.

Ezra: Local legends say it boasts miraculous effects, and can even bring the dead back to life. Obviously, that is likely to be an exaggerated claim.

Ezra: But after examining the related records, we believe the herb's ethanol extract may have powerful neural repair properties.

Ezra: If this can be confirmed, there may be a path to a cure for many presently incurable neurological diseases. Think about the hope that could bring to people around the world.

Thirty-seven: That sounds like a wonderful thing. I didn't know Medicine Pocket was doing such important research.

Ezra: See, you misunderstood, Ms. Mesmer. This isn't just some featherbrained scheme; it's a meaningful project with real value to civilization. That's why I allowed Medicine Pocket to perform their work here in the museum.

The young museum curator is known for his sincere virtue. It is hard to remain indifferent when seeing things through his young eyes.

Mesmer Jr.: But that's still no reason to allow explosions in the museum.

Medicine Pocket: First, don't misunderstand my intentions. I don't give a crap about some miracle cure. I just want to find out if this thing is as amazing as the legends say.

Medicine Pocket: And as for this little incident, you can blame those incompetent Sasquatches responsible for producing the local environment data. Their records are a mess!

Mesmer Jr.: You could have done all this in a lab.

Medicine Pocket: You think I didn't want to? I submitted my application, but that miserable old man's bureaucratic processing has requests piled up to the Foundation's ceiling!

Medicine Pocket: Ophiocordyceps Icarus only grows for a few months. If we miss this chance, we're gonna have to wait a whole year, and I won't wait that long!

Medicine Pocket: So I kicked open the old man's office, and then—

X: And then you got kicked out. I happened to be passing by and witnessed the entire episode.

X: I must concede, Mr. Hofmann had a point: if everyone stormed into his office claiming their matter was more urgent than everyone else's and demanded his immediate assent, there'd be no end to the chaos.

Medicine Pocket: Tsk, that old geezer looks like a wilted gray knotweed, but he's pretty prickly.

X: However, as I recall, he did grant you permission. He said, "Figure it out for yourself."

Medicine Pocket: Exactly. So I figured it out.

Medicine Pocket flashes a mischievous smile.

Medicine Pocket: When you make it big, you have to help out the little guys. That's what good friends do, right? Besides, this museum is huge.

Mesmer Jr.: I maintain my opinion.

X: Anyway, just look, our narrative's protagonists are now all present.

X: When young researchers congregate, we envision it to spark some kind of chemical reaction, a guarantee of a bright future.

X: But there is a minor issue: our two friends here appear to have made a slight divergence in their problem-solving philosophies.

Concern appears on the young curator's face.

Ezra: What happened?

Laboratory, Laplace Museum

Thirty-seven: This is the answer. It doesn't match the numbers you entered into the device, so it went wrong.

Thirty-seven: You need to adjust this number, and you will reach the correct solution.

Medicine Pocket: What? That's impossible!

The draft paper is snatched away. This genius biological researcher who has produced several groundbreaking results in the field of arcanum is no ordinary individual. After a moment, they look up.

Medicine Pocket: Guess what? You were right.

Medicine Pocket: However!

A word indicating a turn, spoken with a particularly firm tone.

Medicine Pocket: You can't just say "adjust a number" and be done with it. The problem isn't solved yet.

Medicine Pocket: How do we actually make that change? This isn't just some adjustable parameter. It's the real world, kid!

Thirty-seven: You mean it's a dependent variable in another function? No problem. Show me the formula.

Medicine Pocket: No, that's not what I mean. Come on, you can't talk about hard numbers and say, "Just change it!"

Thirty-seven: Why not?

Thirty-seven: During my time at the computing center, many people came to me saying, "37, I don't understand why the data is like this. You understand the language of numbers. Tell me what they're saying."

Thirty-seven: They were confused by data anomalies, but there are no "unknown exceptions" or "inexplicable errors."

Thirty-seven: No number is "wrong." It's just that you simply haven't discovered the appropriate theorem to include it.

The girl's face has no hint of pride. She says it as if stating an undeniable fact.

Thirty-seven: If you're confused, then it's because you haven't yet discovered the real pattern.

Thirty-seven: I'd be happy to help, just like I helped the others.

All present in the office hold their breath for a moment.

Medicine Pocket: Wow.

Medicine Pocket: You sure have a big mouth for such a little girl. You think all the puzzles have answers, and you can solve them, just like that?

Thirty-seven: Yes, and no. There are so many puzzles in the universe. Even if the first challengers couldn't solve them, their attempts became the foundation for future attempts.

Thirty-seven: I just think the problem at hand isn't so difficult that we need to leave it to our successors.

Medicine Pocket: Ha! I gotta say ... I haven't come up against a challenge quite as interesting as this in a long time.

The situation seems to be veering toward the chaotic, until someone realizes it is time to intervene.

Mesmer Jr.: Perhaps you two could calm yourselves and be a little more rational in your discussion. Especially you, Medicine Pocket. You know that 37 didn't mean it that way.

Medicine Pocket: What do you mean, especially me? You were standing right here! You heard what she said, right? She started it, so don't go picking on me.

The intervention has little effect, and the conflict seems to be escalating.

Fortunately, there is another participant who has long been skilled at mediating conflict in his own way.

X: Attention everyone! Who would like some ice cream?

All eyes turn toward him.

Mesmer Jr.: You can't be serious, right?

Medicine Pocket: Get bent. Do you think I'm still in kindergarten?

Thirty-seven: What's ice cream?

His initial attempt ran into an unanticipated challenge, but he knows how to regain control.

X: There's a shop in the refreshments area over there.

X: My treat.

The last two words have a predictable effect.

Medicine Pocket: Now you're in for it. I want the full works.

With the first breaking of the ice and the first scoop of ice cream, things become easier.

Dodecagram

DPL-04

Dodecagram

Medicine Pocket: I want to make it completely clear that I'm not asking you to solve that freakin' number problem.

Medicine Pocket: That's my job, and I'm confident I can solve it.

Misunderstandings are cleared up, and cones are bitten through.

Medicine Pocket: Can I just say this? You've got some serious tone issues you need to work on. Not everyone is as broad-minded as me, understand? Hey, are you listening to me?

The answer is no. The girl's attention is rapt entirely on the ice cream being handed to her.

Shopkeeper: Here you go, cutie pie. It's chocolate.

She holds the frosty, sweet treat in her small hand, the refreshing chill brushing past her skin.

Thirty-seven: This ...

Mesmer Jr.: Huh?

Thirty-seven: This is a perfect dodecagram! Why would it look like this?

Mesmer Jr.: You mean? Oh, looking at the shape from above. Right.

Medicine Pocket: Looks like someone's starting up again.

Thirty-seven: I see it! So this is some kind of mold, isn't it? What a tasteful choice!

Thirty-seven: I can see its shape—four equilateral triangles combined—so it can be represented by the Schläfli symbol [12/4 ...

Finding a strange satisfaction with the ice cream's shape, 37 speaks as she eagerly pulls out a notebook and pen, and starts to write.

Both notebook and pen are emblazoned with the Laplace logo. Everyone present has their own, but none have ever placed ice cream and star-shaped polygons side by side like this.

Thirty-seven: Let's start with 12/1—we can't skip it. Although 12/1 can be simplified to 12, so that it's actually a regular dodecagon.

Thirty-seven: Then there's 12/2, 12/3 ...

Five shapes appear on the paper. 37 labels each one with its Schläfli symbol and begins to evaluate them.

Thirty-seven: 12/3 is too clumsy, 12/5 is too extreme.

Thirty-seven: —And the equilateral triangle is the most stable, making the shape it forms the best of the four regular dodecagrams.

Thirty-seven: This is an excellent geometric shape, which means that "ice cream" must be a great dessert!

To express her appreciation of the shape, 37 draws a smiling face beneath the dodecagram labeled 12/4.

The person sitting next to her looks down at the smiling doodle and can't help but laugh.

X: Haha. A commendable choice indeed, but ...

X: Personally, I might favor this rather more charming one composed of three squares.

X pulls out his pen and draws a smiley face next to another shape.

X: I've manufactured gears in this shape before. They proved quiet and reliable in completing some of my more intriguing relays. Unassuming shapes often come with unexpected abilities.

Thirty-seven: I'm a bit surprised you chose 12/3. Squares are solid, simple shapes. They don't match the impression you give off.

X: So, in your impression, Miss 37, I am a restless and complex boy?

Thirty-seven: Well, you certainly aren't a square, but a 12/3 dodecagram is different.

Thirty-seven: It comes from the most basic of shapes but doesn't conform to the same rules. It's unique. I think I've picked up some clues for solving your number puzzle.

Medicine Pocket: Hey, just tell me one thing. Why do you people always aim for the low-hanging fruit? Look, 12/5 is obviously way cooler.

The childish relay continues, and a third smiley face appears, the most arrogant of them all.

Medicine Pocket: Just like an untreated thunder fig husk, prickly like a hedgehog. Its flesh isn't soft or juicy, or even edible, but it's excellent for making anti-electrical paste.

Medicine Pocket: Let me spell it out for you. Only cowards fear shapes; smart people use them.

Thirty-seven: You mean it's challenging, but it holds significance? That makes sense. Each interior angle of 12/5 is 30°, a special case among all acute angles.

Thirty-seven: Its right-angle side is always equal to half the hypotenuse. When we want a right triangle, we almost always choose between an isosceles and this.

Thirty-seven: What about you, Mesmer?

The question comes both inevitably and unavoidably.

Mesmer Jr.: Do I really have to choose one? Isn't this kind of juvenile? Fine, this one.

Thirty-seven: 12/2, a dodecagram made of two hexagons! I thought as much!

Mesmer Jr.: I just picked randomly.

Thirty-seven: But it feels like you. Look at these 120° interior angles. They restrain their sharpness, appearing calm and steady.

Thirty-seven: But hexagons are never truly stable. The collapse of any vertex can lead to catastrophic results. You have two hexagons, which helps, but they lock each other in.

Thirty-seven: Hmm, if either could be enlarged or reduced slightly, that would be an improvement! If they nested perfectly, we would obtain six larger, more aesthetically pleasing isosceles triangles.

Mesmer Jr.: Is this some kind of geometry-based divination? I think it's time we wrap this up.

Thirty-seven: Wait, we have one more shape! The regular dodecagon! We can't omit it just like that.

Medicine Pocket: I'm afraid I have to agree with you on this one, number-girl. I don't think anyone can stand having such an obvious blank space smack bang in the middle of all this.

Thirty-seven: But who could it be?

Mesmer Jr.: Considering where we are, should it not be Ezra?

Medicine Pocket: Well, well. Not so above our "juvenile games" as you implied.

Mesmer Jr.: Look, I just want whatever this is to end as quickly as possible.

Thirty-seven: I'm happy that you've come around to seeing the charm of geometry, Mesmer, but who is Ezra?

Mesmer Jr. draws another little head adjacent to 12, bestowing it with a brighter smile than her own.

Mesmer Jr.: A friend of ours. A very sincere and knowledgeable friend, and the new curator of this museum. I hope you will get a chance to meet him. I think you'd get along well.

Thirty-seven: Hmm. Is he a human then, not an arcanist?

Mesmer Jr.: You've heard of him?

Thirty-seven: No, but you said he's a dodecagon.

Thirty-seven: A dodecagon is different from these other shapes. It's not a star, but a regular polygon. However, a regular dodecagon is indispensable when considering twelve-vertex shapes.

Thirty-seven: Humans and arcanists are different. Their essence differs from ours. They are imaginary, non-existent numbers.

Thirty-seven: But we can still locate them using a coordinate system with i as the y-axis unit. You are a precise fraction, so if you're singing his praises, he must be exceptional. I would really like to meet him.

X: You may need a few days. He's busy preparing a special exhibit. Oh, 37!

With a gasp and a cool, unfamiliar touch, 37 is pulled back to reality from the geometric realm.

Thirty-seven: My dodecagram's vertex is collapsing!

X: It's melting! Quick, take a bite!

Thirty-seven: Nom!

Her little head collides into the ice cream.

Thirty-seven: This ... this is delicious!

Thirty-seven: With such a perfect shape, it was bound to be tasty. But it's still simply astounding!

Medicine Pocket: It's just standard ice cream churned out by a machine, though I'd say it's better than what you can get in the other refreshments areas.

Medicine Pocket: Is this your first time eating this stuff?

Thirty-seven: This ice cream came from that machine?

Thirty-seven: The doctrine tells us not to seek pleasure, nor to indulge desires, but surely bringing this machine back with me wouldn't be a total violation of the rules.

Thirty-seven: It would allow us to focus on studying beautiful three-dimensional shapes more joyfully. 6 would have to approve. Nom! Mmm ...

After some last-second scrambling and some well-placed bites, the melting ice cream disaster is averted.

X: So, have you found any new insights?

Thirty-seven: About ice cream? I've discovered that it's wonderful!

X: No, I meant your research. Have you derived any inspiration from your excursion today?

Her sky-blue eyes blink.

She looks around, then down at the ice cream in her hand, and finally slowly shakes her head.

Thirty-seven: I was not sure.

Mesmer Jr.: Oh, is that so?

Medicine Pocket: Come on! The setup was perfect!

Medicine Pocket: This is supposed to be the part where you realize you discovered some profound truth in the mundanity of this crappy ice cream, shout "eureka," and then rush back to the lab to work on your breakthrough.

Mesmer Jr.: Medicine Pocket, you don't really believe that, do you? No one at Laplace would laugh at that joke.

Thirty-seven: The exhibits in this museum are all very interesting, and this ice cream is delicious. I've found many good numbers today.

Thirty-seven: But ...

The girl looks down at her hand. Where her perfect dodecagram had been, but now nothing remains.

Then she looks up.

Upon the real-time star map on the museum ceiling, stars twinkle beautifully. A rare chance to appreciate the grandeur of the cosmos with the naked eye.

The paths of the planets are clearly marked, and the vast Milky Way is unobstructed by light or atmospheric pollution. Every pixel is perfectly placed.

Thirty-seven: What I do at the computing center seems to be the same. I came to get more numbers, talked with people from different fields, and it brought me these in return.

Thirty-seven: I think I've grown somewhat. I've seen things, learned more, and gained more numbers than when I was on the island.

Thirty-seven: However ...

Thirty-seven: When I look down, I still can't see where the next square is.

Mesmer Jr.: Oh, is that so?

This is not the expected result of her treatment.

Mesmer Jr. suddenly recalls the conjecture about planetary distances and Platonic solids that the girl once marveled at in the lab.

Johannes Kepler, hailed the "legislator of worlds," discovered the three laws of planetary motion. In his book, Mysterium Cosmographicum, he proposed a hypothesis.

He claimed that the distances of the six planets that had been known at the time—Mercury, Venus, Earth, Mars, Jupiter, and Saturn—from the sun could form five perfect Platonic solids.

The arrangement fit so perfectly. Deep down, even the atheists of the time hoped that God had arranged the universe according to such a delicate blueprint.

But as we know now, his hypothesis was incorrect.

Mesmer Jr.: ...

Mesmer does not fully understand 37's goals or her methods. But she astutely observes that 37's research and all her efforts are predicated on a shared hypothesis.

A hypothesis as beautiful as Kepler's conjecture, but equally untrue.

Mesmer Jr.: Have you considered ...

That your hypothesis might be wrong?

Thirty-seven: What?

Mesmer Jr.: Nothing.

Thirty-seven: Perhaps I was a little hasty.

37's gaze shifts away from the invisible square on the floor.

Thirty-seven: John Napier took twenty years to devise logarithms, and Fermat's Last Theorem took 358 years to be fully proven. It's likely that my answer won't be found overnight, either.

Thirty-seven: I know that I need to hurry. I can't afford to waste time. But ...

Thirty-seven: The doctrine instructs us never to lose composure or reason, to adhere to the laws of all things and ascend the divine steps, so we can reach the highest height, and not descend into the abyss of chaos.

Thirty-seven: I shouldn't forget my limitations.

The other three exchange glances.

Medicine Pocket: Congratulations! It seems our mathematician has finally discovered a basic truth.

Medicine Pocket: Great achievements don't come overnight. The more challenging your research, the more time and effort it will take. These things are proportional.

X: Your research is fascinating, and I look forward to the day it is completed, but don't overexert yourself. That's just a little bit of advice from your senior at the computing center.

X: Many talented researchers have focused excessively on their work, forgetting they too are mortals. We don't want you to follow in their footsteps, 37.

Thirty-seven: Hmm? I don't know if I understand what you mean. I'm feeling just fine.

Mesmer Jr.: I thought as much.

X: Wow, this is worse than I anticipated. Look ...

X chooses his words carefully, but Medicine Pocket just rolls their eyes and chimes in.

Medicine Pocket: You look like a deflated bloodmoon jellyfish, oozing a vibe like you've been stuck in a plastic bottle without a water change for days. You've got tentacles floundering all over the place!

Medicine Pocket: And I've been meaning to ask: What's with the messy hair? Even the cactus cat in my lab grooms itself better than you.

37 looks around in confusion, attempting to locate her fictitious tentacles floundering in the air.

Thirty-seven: Oh, I see. It's a metaphor, isn't it?

Medicine Pocket rolls their eyes again in response.

Mesmer Jr.: Whatever else you do, try not to stay cooped up in that lab for too long. People need a little entertainment to keep themselves sane, and that's advice from a therapist.

Mesmer Jr.: I already have enough work on my hands. I don't want to see you lining up at the door of the artificial somnambulism therapy room.

37 blinks, contemplating for a moment, before fixing her bright eyes on the group.

Thirty-seven: Are you all really so concerned about me?

Someone takes issue with her choice of words.

Medicine Pocket: Yeah, right! Like how nuts are we for caring if you work yourself into madness?! You're already beyond delusional.

Thirty-seven: You're all so kind. Thank you.

Thirty-seven: I don't feel like I'm pushing myself, but spending time with all of you makes me happy. It reminds me of the past.

Thirty-seven: Whenever my friend Sophia would come to see me, if I was working on a problem, she'd linger at the door for the longest time before coming in.

Thirty-seven: She'd always ask, "Am I disturbing you, 37?"

Thirty-seven: To tell the truth, her visits did sometimes slow down my progress.

37 shakes her head.

Thirty-seven: But research can always be continued later. The truth is, I was happy she came to see me.

X: It sounds like you have a good friend there.

Thirty-seven: Yes. She ... She's not here now ... but I hope I can introduce her to everyone someday.

X: I'd be honored to share this secret base of ours with you and your friend.

Their time together seems to be nearing its natural close.

X: Speaking of which, remember the special exhibit I mentioned?

X: It opens in three days. Our friend Ezra—our dodecagon—exerted considerable effort to facilitate it. Medicine Pocket and I assisted him a little too.

X: If you have time, you'd be very welcome to come and join us as we view its offerings. We can peruse the exhibit together. Of course, if our dear therapist would like, she can join us, too.

Thirty-seven: Thank you for the invitation. I'd like to come. Then I can have another ice cream.

Medicine Pocket: You really like that junk, huh?

Thirty-seven: I just appreciate its geometry. It's nothing related to its taste. I think it helps my mind work more efficiently.

Medicine Pocket: That's the lamest excuse I've ever heard! So if I gave you a scoop of anti-electric paste that looked like ice cream, would you eat that too?

Mesmer Jr.: Give it a rest? Don't you find this line of questioning to be a little childish?

Medicine Pocket: Why are you always taking sides? And worse, never mine!

Their farewell chatter seems to go on without end. The young researchers debate over the essence of things, and the functions of the taste and digestive systems.

Until, at last, the fifth party present brings an end to their interminable digressions.

Shopkeeper: Uh ... sorry, everyone, excuse me.

Medicine Pocket: Huh?

Medicine Pocket seems to have genuinely forgotten the ice cream shopkeeper was there.

Shopkeeper: It's just, I heard you say "next time," and, I'm glad you enjoyed yourselves, but ... I'm afraid there won't be any more ice cream next time.

Shopkeeper: My shop, well, it's closing down.

Logarithm

DPL-05

Logarithm

After the owner announces the closure, a brief silence ensues.

Thirty-seven: Really?

Medicine Pocket: Closing down? What do you mean "closing down"?

Shopkeeper: Uh, I mean, I'm going out of business. I can't keep this shop afloat.

Medicine Pocket: Of course, I know what "closing down" means. Ugh ... Do you really think I need you to explain THAT to me?

The questioner rudely interrupts.

Medicine Pocket: Don't tell me that this means I have to go back to Clever Cones? Their Carbuncle servers always sneak bites from my order! When I complained, they said it was within "normal loss allowance"!

Thirty-seven: So, we won't be able to have this ice cream anymore?

The girl who was about to leave stops in her tracks.

Thirty-seven: That beautiful, regular dodecagram, and its magical, refreshing, and rich flavor ... The next time I come here, I won't be able to taste it?

Thirty-seven: Happy, beautiful things, never to be seen again. What terrible news.

37 thinks for a moment.

Thirty-seven: I must admit, I don't know much of anything about economics. But if this shop sells such beautiful ice cream, yet it's closing down. There must be something wrong.

Thirty-seven: How about this? I'll help you solve your problem.

It appears their farewell was premature.

Thirty-seven: So, then I believe this is the problem.

After subjecting the shopkeeper to a dizzying series of questions, the little nascent consultant consumes another free ice cream as she announces her solution.

Thirty-seven: This chart illustrates the gap between your shop's data and that of other shops. Apparently, your foot traffic and average transaction value are significantly lower—that is the issue.

Thirty-seven: So, we only need to increase these two numbers, and your troubles will be over.

Shopkeeper: Really? Uh, which ones did you say?

The owner leans over the charts, attempting to decipher the many points and lines, growing more and more confused.

Medicine Pocket: Are you serious? I feel like I've walked onto the set of some boring sitcom.

Medicine Pocket: This is my private lab—a place for ideas of consequence! Now you've barged in here with your kitschy number name and festooned my brainstorming board with profit-loss analyses for a two-bit ice cream stand!

Mesmer Jr.: Correction: This is not your private lab. But I admit, I am questioning why we're still here, too.

X: Come on, friends. We all know why. We're having a good time, aren't we?

Medicine Pocket: You wouldn't know fun if it came up to you and sat on your head.

Medicine Pocket: "Let's all sit around having 'fun' while we discuss how an ice cream shop owned by some idiot that doesn't even understand business is failing ... Duhhh!"—the answer's obvious.

Medicine Pocket: How can you expect to help someone who, when asked, "How much product do you need to restock daily," answers, "I don't know exactly, but it feels like a lot"?

Shopkeeper: Hey! I've put up with your insults long enough! I can hear you, you know?!

Shopkeeper: You're the one who's always making unreasonable requests, like asking for refills when you've eaten all the ice cream off your cone. To think I actually topped you up, and now you're making fun of me?!

Medicine Pocket: I'm just stating the facts. People can excel at some things but suck at others.

Medicine Pocket: Your ice cream is to die for, but your business is a mess. Why don't you just sell your recipe?

The shopkeeper is speechless, her face slowly reddening.

Medicine Pocket: You can't be serious.

Shopkeeper: Fine, I admit it! This is my dream! To own an ice cream shop—it's always been my dream, and I don't want to give it up!

The owner turns away from Medicine Pocket's rolling eyes and faces her would-be savior.

Shopkeeper: I was ready to give up, but then you appeared out of nowhere. Do you think you can really help me, little girl? What was it that you said I need to increase?

Thirty-seven: Average transaction value and foot traffic. Look, your average transaction is about a third of other shops.

Thirty-seven: And your foot traffic is only 63 people a day, which is just not enough. That's why you're losing money: your revenue is too low.

Shopkeeper: I thought 63 people was a lot! Think about it—over sixty people!

Medicine Pocket: That's some scary level of obliviousness, lady. Look at the other stores around you! They've got lines of people queuing up at Clever Cones' ice cream truck every day! That's hundreds and hundreds of ice creams daily!

Medicine Pocket: You should thank Ezra for not using the museum store to make a buck, or you'd be losing even more. He lacks business sense, like someone else I know.

Shopkeeper: Okay, okay! You're right.

The shopkeeper surrenders to their tirade, reexamining the chart 37 drew.

Then, she does something almost unheard of at the Laplace Scientific Computing Center: She questions 37's conclusion.

Shopkeeper: But I sell ice cream, and the others sell souvenirs, so of course their average transaction value is higher.

Shopkeeper: I can't price my ice cream to match their average transactions. I know it tastes great, but honestly, I don't think people will pay that much.

Thirty-seven: You mean this ice cream isn't worth a higher price? Even with such a beautiful regular dodecagram shape?

The mathematician, unfamiliar with prices, is momentarily stunned, then humbly accepts this correction.

Thirty-seven: Then we only have foot traffic left to work with. We will have to increase that number.

Thirty-seven: Just raise this number to 2.56 times the original, and we can break even.

Thirty-seven: Although I don't particularly like this variable—2.56 isn't an exact number; it's merely a decimal approximation.

Shopkeeper: 2.56? Uh, multiplying that by the 63 people entering the store, that would be ... uh ...

Thirty-seven: You see, this is the downside of using approximations. You'll get 161.28, but the number we're discussing is actually 161.

Thirty-seven: And that's a bit counterintuitive. 161 is a deficient number; its proper divisors add up to less than itself.

Thirty-seven: I think once you reach the goal of 161, you might pursue better numbers, like its next palindrome, 171. Then 181, 191, 202 ...

Shopkeeper: Wait, wait! Please, just stop for a moment.

Although she doesn't understand the girl's calculations, the owner notices the numbers are getting bigger.

Shopkeeper: Let's start with our first goal. You want me to sell 161 ice creams a day? Is that even possible?

Shopkeeper: That's a whole lot of people! How can I serve that many customers every day?

Thirty-seven: There's no need to worry. Your business hours are from ten in the morning to six in the evening, so you only need to sell one ice cream cone every 2.98, or rounding up, every 3 minutes.

Thirty-seven: I didn't time you when you served us, but I simulated the process, and if everything goes smoothly, I think you can do it within 30 seconds.

Thirty-seven: It should certainly be possible.

The shopkeeper's eyes drift into the distance, imagining all the customers.

Shopkeeper: I-I don't know. Maybe I can try?

She's convinced. But then—

She asks the crucial question.

Shopkeeper: But, how do I get these 2.56 times more people to come to my store?

Thirty-seven: Let me think ... Hmm ... hmm ...

The talented mathematician opens her mouth, then closes it.

After a while, she finally tries to give an answer.

Thirty-seven: Multiply by 2.56, then subtract 0.28?

Shopkeeper: Huh?

Medicine Pocket: Haha! What did I say? You can't just say "increase the number."

The long-silent onlooker seizes the opportunity to spit a little venom.

Medicine Pocket: Welcome to the real world, Number Brain.

X: This might be quite sudden, but I think it's time for a drastic reappraisal of our understanding of mathematics.

X: Many people find the world of mathematics to be too abstract and difficult to comprehend, and so they're afraid of it.

X: You can't blame them. Think about an apple, a person, an event, a success, or a failure. These concepts are easy to grasp.

X: But when numbers first detached from their counted objects, when "1" became "1"—and just "1," mathematics departed from the material world we are all familiar with.

X: Bound by our physical forms, we can only chase those numbers in abstract. We study and research mathematics, using those numbers to borrow power from that transcendent world.

X: In that world, "nothing" boldly appears as 0, negative numbers stand on the opposite end of "existence," and even "infinity" can be included in formulas ...

X: In the world of mathematics, we can do anything—at least, theoretically. Our minds are free, no longer constrained by reality.

X: Only a few understand this freedom's charm, and fewer still wield true omnipotence in mathematics. Like our delightful new friend here.

X: But there's just one small problem.

Mesmer Jr.: ...

The silent audience stares at the eloquent researcher, then coldly looks away.

Most people would feel awkward and stop talking, but this unknown number is no ordinary digit.

X: Do me a favor, will you? Ask me, "What problem?"

Mesmer Jr.: ...

Merciless silence.

X: The problem is, mathematics has the power to do anything in the abstract, but to make it work in reality? For that, we need to follow three steps.

X: First, we need to abstract our real-world problem into a mathematical one.

X: Second, we need to solve this problem within the mathematical world.

X: And third, we need to find a way to bring that abstract answer back to reality.

Mesmer Jr.: You mean ...

His audience understands. Mesmer Jr. looks outside.

Thirty-seven: Past data shows an average of 9,837 visitors per day during this quarter over an eight-hour operational period, so after three hours, about 3,689 people should have entered.

Medicine Pocket: Hey, you know visitor numbers vary by time of day, right? There's a "peak period" … I thought that was just common sense.

Thirty-seven: Even so, the gap is too big. Only 457 people reached an area where they could see the shop's sign.

Thirty-seven: People aren't coming to the refreshments area where the ice cream shop is located. That's the root of our problem!

Shopkeeper: Huh? Four hundred people sounds like a whole lot!

Medicine Pocket: You ever hear of an entry rate? Not all four hundred of them are hungry, so they don't all walk in! And only twenty that do will buy anything!

Medicine Pocket: So only about 12% of visitors come through this refreshments area. I have to admit, that's lower than I expected.

Medicine Pocket: No wonder I like sitting here. There are hardly ever any annoying people around to bother me.

Thirty-seven: But why don't people come this way? I don't understand. Wait, this map …

The brain that solved the computing center's year-long sales problem in seconds is now buried in a small puzzle about ice cream and visitors.

Mesmer Jr.: She's on the third step.

Returning to reality.

Thirty-seven: I've got it!

Fractal

DPL-06

Fractal

Thirty-seven: Numbers tell us everything.

At the round table, 37 gazes at the beautiful dodecagram treat in her hand, then at the long line of visitors outside the ice cream shop.

Thirty-seven: We graphed the foot traffic in every museum hallway, the composition of people walking through them, and compared the impact of other related facilities on foot traffic …

Thirty-seven: And then we found the coordinates of these numbers in the real world.

Thirty-seven: The refreshments area housing the ice cream shop is in the passage connecting exhibits B3 and C2, but this passage is narrow, so most people visit all of Zone B, then take another route to get to Zone C.

Thirty-seven: People looking for a refreshments area feel like they have better options—the hallway to the office area takes space, so this refreshments area also lacks a restroom.

Thirty-seven: Visitors with children are important customers for an ice cream shop, but this refreshments area is in the opposite direction of the play area.

Thirty-seven: And so on. I've found what's behind the numbers, the hints they've given us.

Thirty-seven: Behind the numbers are people. Their habits, thoughts, and subconscious actions lead to this result.

Thirty-seven: Honestly, it feels strange and … counterintuitive. It's completely unlike my usual way of observing the world.

Thirty-seven: I always observe the phenomenon first, then look for the numbers behind it, not the other way around. I don't know if it'll help my research, but …

Thirty-seven: I see them now.

Seeing his new friend daydreaming again, X shrugs and continues.

X: In short, 37 told us the answer behind the numbers. The subsequent steps were elementary.

X: Our key issue was the shop's disastrous location, so we needed a robust publicity campaign to direct people to our hidden gem!

Medicine Pocket: Actually, linking the marketing with the special exhibition was my idea. Ezra, if there's a prize going to the hero who brought more visitors to your exhibition, don't forget who's responsible.

X: And our therapist helped as well. She suggested some new ice cream flavors. See those orange bits? They're candied lemon peel.

Mesmer Jr.: I just heard people around the office mention they like this kind of thing.

Thirty-seven: It was a great idea, Mesmer! They're like lucky little numbers scattered on a plotted line!

Ezra: And thus, the "Star of Wonder" ice cream was born.

Ezra takes the last bite, feeling the refreshing ice cream melt in his mouth, waltzing with the crispy cone.

Ezra: It's delicious, really. And your story has been so inspiring.

Ezra: Before you came along, the only people that would visit the exhibitions here were fellow researchers in similar fields. But thanks to this ice cream, we've attracted more people than ever!

Ezra looks behind him.

Among the streams of visitors are scholars, locals, and tourists who seem to have traveled from near and far, flanked by young kids.

Some talk passionately in front of the exhibits, some quietly read the descriptions, bursting up with sudden realizations, and others simply admire the beautiful displays.

But they all share happy smiles, and every pair of eyes looking at the exhibits lights up.

Ezra: I've been feeling a bit nervous. Today is my first public speech since being appointed curator of the museum.

Ezra: I've tried some tips to help ease my nerves, like imagining the audience as potatoes or mushrooms. But I think something you said will help most, 37.

Ezra: Behind the numbers are people. They came from all over for the special exhibit. They're real people with a real love for knowledge. They're not potatoes or mushrooms.

Ezra: That's the real trick to this.

Thirty-seven: Is your speech about to start?

Ezra: It is. Would you do me the honor of joining my audience?

Thirty-seven: Of course! I'd love to!

The young researchers rise from the round table and escort their friend to his proving ground.

Thirty-seven: Wow, there are as many people here as in our lecture hall.

Ezra: Oh no, I'm losing my nerve again. Maybe I should try imagining everyone as mushrooms? No, that's avoiding the issue, and it's rude.

Thirty-seven: Why would you be nervous? No matter who listens or how many people there are, the content of your speech remains the same, doesn't it?

Thirty-seven: The Truth doesn't change, and that's all we should be accountable to.

Ezra: You're right. Miss 37, very right. We worked hard to get this Miracle Stone ready for display. I can't sell it short. Phew!

X: You must exhibit the same spirit you demonstrated while ascending that snow-capped mountain, my friend. Your task today is trivial in comparison.

Medicine Pocket: If you need to, keep an eye on us; I have ways to keep you too busy to be nervous.

Mesmer Jr.: You'd better not have any bad ideas running around that mind of yours. Go on, Ezra, imagining everyone as mushrooms can be your fallback plan.

Ezra: Thanks, friends. I'm on now!

As the young curator steps up to the center of the stage, the murmurs surrounding him subside.

Ezra: Welcome, everyone! I sincerely thank you all for coming.

Ezra: I haven't been the curator of the Laplace Museum for long, but I'm honored that my call has been answered from all over the world.

Ezra: You have different identities, live different lives, have different passions ... but our wishes and ideals are the same.

Ezra: That is, to spread knowledge, to share experiences, and to let the vastness and beauty of the world be seen by all.

Applause ring out. The young curator has earned their respect and trust through his remarkable achievements.

People listen to him with excited murmurs, already anticipating the surprises he will unveil.

Ezra: Many precious exhibits have been sent to the museum, their owners generously lending or even donating these treasures, so their splendor can be appreciated by all our visitors here.

Ezra: This number far exceeded my expectations. I must admit, while it's an honor, the extensive appraisal work has been a challenge for us.

Ezra: But fortunately, with the help of many knowledgeable friends and colleagues, it was a challenge we overcame. I am truly fortunate to have such passionate and considerate companions.

The young man in the center of the exhibition hall looks toward a corner of the venue, smiling and nodding.

Medicine Pocket: Hmph, no need to thank us, although we did do a good job, didn't we?

Mesmer Jr.: Quiet, you.

Medicine Pocket: Hey!

Thirty-seven: Haha, you're all too funny!

The whispers draw a few eye rolls, and the young researchers quiet down. The speech continues.

Ezra: All the donated specimens we've received are invaluable. Each one is unique.

Ezra: But among them, we unanimously believe this piece holds the most remarkable significance, and for that reason we created a special exhibition to highlight it.

Ezra extends his hand, directing everyone's gaze to a pale blue, semi-transparent crystal placed at the center of the stage.

Ezra: The Miracle Stone—from an ocean of crystals that cover Neptune's surface.

Murmurs and exclamations of wonder rise as people gaze at the beautiful crystal.

Ezra: Structurally, it is a diamond. In fact, it is not much different from the cut diamonds we know.

Ezra: Instead, its uniqueness lies in the fact that this beautiful stone comes to us from 4.5 billion kilometers away.

Ezra: As with previous attempts, Voyager 2 carried an arcanum space teleportation formula—an attempt to obtain material samples from other planets.

Ezra: But as many know, these previous attempts had failed.

Ezra: Although arcanists have always suggested that spatial distance was not the true reason for the formula's failure, ultra-long-distance space teleportation was still considered impossible.

Ezra: Unfortunately, during the 1980s, space agencies worldwide stopped equipping their probes with these arcanum formulas to lower costs.

Ezra: Yet, we still had a chance. When Voyager 2 flew past Neptune, it succeeded where the others failed, bringing us this gift from a distant planet.

Ezra: Many mysteries surrounding this beautiful diamond from afar remain unsolved.

Ezra: We all know Voyager 2 reached its closest point to Neptune on August 25, 1989, but this diamond was teleported back much later.

Ezra: Is the arcanum ultra-long-distance teleportation formula so different from ordinary teleportation that it causes a noticeable time delay? For now, with only one case example, we can only speculate.

Ezra: Still, according to the space center's report, this beautiful stone appeared, and the complex and precise formula appears to have finally succeeded.

Ezra: At the same time, the detection system also found arcanum traces not belonging to the original formula. They left a faint, almost imperceptible fluctuation before disappearing, and we haven't detected any similar arcanum fluctuations since.

Ezra: Several researchers claimed to have heard a beautiful melody when the Miracle Stone appeared. The melodies they hummed were astonishingly consistent, making it hard to believe they simply hallucinated the sound.

Ezra: However, we couldn't find any recordings matching their descriptions on any of the space center's detection devices ...

Ezra: With so many uncertainties surrounding this stone, people often ask—is it real?

Ezra: Could this really be a diamond from Neptune? From the second-farthest planet from the sun in our solar system? What secrets does it hold?

Amid the expectant gazes, the young curator shakes his head.

Ezra: Of course, we have no definitive answers for now.

Ezra: Voyager 2 still sends signals to Earth, but aside from the miraculous "teleportation successful" data written into its records, it cannot elaborate on its achievements.

Ezra: The only certainty is that this diamond indeed appeared at the teleportation circle's reception point.

Following his gaze, the audience focuses again on the beautiful crystal in the center of the room.

Ezra: The space center researchers were extremely careful, and after finally confirming it posed no danger, the Miracle Stone was prepared for exhibition, allowing us all to behold its brilliance.

Ezra: It is a miracle, jointly brought to us by the union of science and arcanum, and a wondrous mystery waiting for us to unravel.

After a hard-fought battle, the young curator returns to the refreshments area, surrounded by his friends. They sit down again at the round table.

Ezra: How did I do?

Ezra: I wrote my speech out ahead of time, but I accidentally skipped over a lot of parts when I actually got up there on the stage.

X: Your concerns are unwarranted, Ezra. Your exhibition and your speech received universal acclaim.

Thirty-seven: I don't know much about astronomy, but I found your speech to be inspiring. You remind me of a friend from my home.

Thirty-seven: He's a bit like you, often quiet and reserved, and never keen on being the center of attention, but when he speaks in front of a crowd, everyone stops what they're doing to listen.

Thirty-seven: Oh! I almost forgot!

37's exclamation draws looks from her friends, except for Mesmer Jr., who quickly realizes what she means.

Mesmer Jr.: Your letter.

Thirty-seven: I haven't written back to 6 yet! Let me see. 6 won't mind if it's a little rough. I think I can tear a few pages from this notebook and use it as letter paper.

X stops her before she can harm her notebook.

X: Hold on now. I recall there's some Laplace-issued stationery and envelopes in Medicine Pocket's room. Why don't you use them?

X: It is better to receive a letter with respectable presentation.

Medicine Pocket: What's the big idea—promising people my stuff without asking me? It's not that I mind, but do I even have any of that stuff?

X: You do. I saw them in your, I want to say, pile of junk? Come on. Let's retrieve them together.

Medicine Pocket: Did you just call my office a "pile of junk"?

The two walk off, bickering. The kind curator watches them with concern.

Ezra: Uh, they won't fight, right? Oh, no, they're already arguing. Maybe I should follow.

Desert Flannel: Found him. He's over here.

Spathodea: There he is! Hey, Ezra!

Ezra: Spathodea! And Desert Flannel too! What brings you here?

It looks like the young curator has other guests.

Spathodea: I missed your exhibition last time because of that fight, so I thought I'd surprise you today!

Spathodea: And wow, I mean, there were so many people! Such a big event, and you're the host. You're amazing, Ezra!

Desert Flannel: We just listened to your speech. Very impressive, little curator.

Ezra: Thank you! I'm so glad you came.

Friends always have endless things to discuss after a long time apart. Ezra finally takes Mesmer Jr.'s advice, apologizes, and leaves to greet his friends from far away.

The real-time star map on the ceiling still shines brightly, and only the therapist and the girl favored by numbers remain at the round table.

Mesmer Jr.: What do you think of the diamond from Neptune?

Thirty-seven: Why are you asking?

Mesmer Jr.: You haven't expressed your opinion on it, and you often have a different perspective from most people. I suppose I'm curious to know where you stand on this "Miracle Stone."

Thirty-seven: It's beautiful, like the garnet we saw that day, with the same orderly, solid structure.

Thirty-seven: Though I don't quite understand why people favor diamonds over other isometric minerals, or why this one is considered more special than any other.

Thirty-seven: Space and time exist only in the phenomenal world. An object's essence isn't determined by where or when it originated.

Mesmer Jr.: So, you don't think all that much of it?

Mesmer Jr. recalls a word used on the promotional poster.

Mesmer Jr.: Maybe besides its astronomical significance, people admire this stone for the "fantasy" it represents.

Thirty-seven: You mean because it comes from far away, people value it more than similar items from much closer? That doesn't seem fair.

Mesmer Jr.: For most people, yes. Emotional preferences lead us to make less objective judgments.

The girl appears contemplative.

Thirty-seven: Like how people didn't come to this ice cream shop. Those were also emotional preferences, fleeting thoughts, not solid, objective reasons.

Thirty-seven: Honestly, I don't really understand it.

Though the girl solved the ice cream mystery with numbers, she still doesn't truly understand the people behind them.

It isn't her fault. A bird is destined to never understand the feelings of a fish. If others are all trapped in a cave, this is the girl that is born to see the vast starry sky on the other side of the rocky walls.

Asking her to focus more on the stifling cave around her would be asking too much.

Mesmer Jr.: Fine.

The therapist decides to give up. Just as she has been telling herself from the beginning. This is none of her concern.

But suddenly, she recalls a story.

A story she heard long ago, when she was at the SPDM with her old friends.

Mesmer Jr.: 37, do you know how Neptune was discovered?

Thirty-seven: Huh?

A Logarithmic Axis

DPL-07

A Logarithmic Axis

Thirty-seven: Mesmer Jr. told me the story of Neptune.

Thirty-seven: Using the elegant formula of gravitational theory, people discovered Uranus. But they found that Uranus didn't follow its calculated trajectory.

Thirty-seven: They began to wonder: Was the theory wrong? Did they fail to account for something?

Thirty-seven: Some people chose to deny it, others to avoid it, but some decided to face the challenge head-on.

Thirty-seven: They wanted to prove that what they believed in was correct, so they made a hypothesis: what if gravitational theory wasn't wrong?

Thirty-seven: What if the deviation in Uranus's trajectory was caused by another, as yet undiscovered, eighth planet?

Thirty-seven: Proving this wasn't too difficult. Astronomers only needed to point their telescopes in the right direction, following the calculations precisely.

Thirty-seven: And then they discovered it, Neptune.

A Reply from Across the Waves: I really liked that story. I think now I understand what people mean by "fantasy."

A Reply from Across the Waves: People often make decisions that are difficult to understand—decisions based on sentimental likes and dislikes. But behind those seemingly irrational decisions, there may actually be logic.

A Reply from Across the Waves: Mesmer said that people have things they love and things they believe in, so when they find something in the phenomenal world that aligns with these things, they feel happy.

A Reply from Across the Waves: I think one way of understanding it is: people make their own hypotheses. When those hypotheses are proven, when they find results that fit the rules, people give them special meaning.

A Reply from Across the Waves: That meaning is invisible, intangible. In the same way, numbers exist in everything—untouchable but real.

A Reply from Across the Waves: They all connect to the transcendent world we long for.

A Reply from Across the Waves: We make our own hypotheses, too. Sometimes we can prove them and discover a new theorem; sometimes reality tells us, "No, you're wrong."

A Reply from Across the Waves: I'm glad that my ice cream didn't say "no" to me.

A Reply from Across the Waves: This project opened my eyes to something.

A Reply from Across the Waves: I took my problem from a logarithmic axis to the addition axis, and turned my numbers into corresponding addends, so now ...

A Reply from Across the Waves: I should take my sum back to the original axis.

A Reply from Across the Waves: It's such a simple answer. If only I would have thought of it sooner.

A Reply from Across the Waves: Still, I think I'm starting to see the outline of my next square.

Thirty-seven: After that, whenever my research progress stalled, I would go to the museum to relax. It's not as ideal as the beach, but it's nice in its own way.

Thirty-seven: The ice cream shopkeeper is still very grateful to us. Every time I go, she tries to treat us to free ice cream.

Thirty-seven: But I've done a lot of jobs for the researchers at the computing center, and I've earned enough points to pay for myself and treat everyone else now.

Thirty-seven: The shopkeeper stopped making the "Star of Wonder" flavor after the special exhibition ended, but I don't mind. My favorite flavor is chocolate anyway.

Thirty-seven: Her ice cream is really wonderful. Its beautiful geometric shape helps me organize my thoughts.

Thirty-seven: So every time I go to the museum, I choose to eat just the right amount of ice cream.

Shopkeeper: Don't be shy. Eat as much as you like! There's plenty more where that came from!

Medicine Pocket: Don't you worry about that. I never pass up a freebie.

X: And what does this prove? That good things happen to good people.

Mesmer Jr.: Really, you two? You don't feel any shame over accepting 37's ice cream benefits on her behalf?

Medicine Pocket: I know I don't.

Mesmer Jr.: Of course, you don't.

X: But more seriously. It's not recommended to consume ice cream in large quantities. Be careful not to get a stomachache, 37. If I'm not mistaken, you've had more than one cone already.

Thirty-seven: Don't worry, this is only my third cone today.

Thirty-seven: 1 and 2 make 3, representing the virtues of moderation and prudence. 3 ice creams couldn't possibly be too much.

Thirty-seven: Though perhaps 4 is a better number. It's the smallest composite number among positive integers, the "greatest miracle," the "most perfect number"! Clearly, 4 is a better final answer.

Mesmer Jr.: You mean you want another one? I don't recommend it.

Thirty-seven: Why not? 4 is obviously better than 3.

Medicine Pocket: I think 10 is better, myself. It's the smallest two-digit number in the decimal system, and when you write it in the ten's column, it only counts as a 1! How cool is that?!

Medicine Pocket: But just a friendly reminder: this refreshments area doesn't have a restroom.

Thirty-seven: I often see my friends, Mesmer Jr., X, Medicine Pocket, and Ezra there. They're all fascinating people in their own rights.

Thirty-seven: Being with them often reminds me of the times we spent with Sophia, and even with 210.

Thirty-seven: Sometimes we see Spathodea and Desert Flannel. They're Ezra's friends from far away.

Thirty-seven: Spathodea is a positive integer and always full of energy, while Desert Flannel is a bit like Mesmer Jr.—a fraction who can be strict with herself at times. They're fascinating too.

Thirty-seven: But then one day, we received bad news. Terrible, heartbreaking news.

Thirty-seven: All things in the phenomenal world are uncertain and unstable. I have known this truth for a long time, but it still came as a surprise.

Shopkeeper: Hey, sorry to interrupt you.

Shopkeeper: I just wanted to let you know, the ice cream shop's closing ... again.

Medicine Pocket: Huh?

Medicine Pocket: What nonsense is this now? Look at this long line. Are you really telling me you can't even manage with this much success?

Shopkeeper: No, it's not that I need to close. But ... I did some math, just like 37 taught me.

Shopkeeper: I've been making good money recently, but once this rush is over, my revenue won't be as high. Even with decent daily earnings, it would take over two years for me to break even.

Shopkeeper: So I thought, why not hand the shop over, like someone did to me? If I sell it while it's hot, I can recoup my investment.

Thirty-seven: It sounds like you've changed your thinking and improved your reasoning skills. I guess that's ... a good thing.

Thirty-seven: But if the shop is closing, soon, I'll have no ice cream to eat.

Thirty-seven: I'll never see those beautiful dodecagrams again, which is definitely not a good thing.

Thirty-seven: But that's not the worst part.

Medicine Pocket: So, that's it? After all we went through, I'm still gonna have to live life without these cones? What happened to your dream?!

X: Regardless, I must say, it's a very rational decision. Don't worry, ma'am. I support you fully.

Mesmer Jr.: And if this unreasonable customer tries threatening you, you can contact the curator over there.

Medicine Pocket: Hey, Ezra! Is this allowed? Do something!

Ezra: Transferring ownership of the shop isn't a violation of her contract. I can't abuse my power on this one.

Medicine Pocket: What will I eat when I go for my walks? Those visitors passing by? I bet they'd taste terrible! Anyway, what's the new owner gonna sell here?

Shopkeeper: I heard he's going to sell beans in all kinds of weird flavors. You know, like crab, Scale of Demonic, snail eyeball ...

Medicine Pocket: Who would want to eat snail eyeball-flavored beans?!

Thirty-seven: But we can't eat beans!

Medicine Pocket: Huh?

Medicine Pocket: You don't mean that eating beans is against your religion, do you?

Thirty-seven: Of course, it is! Eating beans is a terrible sin—the worst sin imaginable!

Thirty-seven: If you eat beans, you'll be sentenced to death.

Ezra: That's ... Sorry, no offense, but that sounds a bit harsh.

Thirty-seven: Considering you're not from our school and we're not on the island, you won't get punished.

Thirty-seven: But I can't eat them all the same.

Thirty-seven: What a shame! A shop mutated from selling beautiful ice cream into evil beans.

Medicine Pocket: Hold on, but you're eating chocolate ice cream! Isn't that the same as eating beans?

Thirty-seven: What?

Medicine Pocket: Chocolate ice cream is made from chocolate, and chocolate is made from cocoa beans. Cocoa beans are beans.

Medicine Pocket: Therefore ...

Medicine Pocket: You've been eating beans.

Thirty-seven: ...

Thirty-seven: No.

Thirty-seven: It's impossible.

Thirty-seven: Impossible! If I really ate beans, then ... then I should have been taken by an Abraxas by now!

Thirty-seven: But I haven't been taken by an Abraxas! This result is contradictory, so there's only one possibility: the hypothesis is incorrect! You must be lying to me!

Ezra: 37, your ice cream's melting! I brought a handkerchief. Here, you can wipe it up.

X: Save the poor ice cream! And don't worry. They're not beans.

Medicine Pocket: You really believe that? You're gonna believe this guy? Hah! Maybe your Abraxas is simply on its way.

Mesmer Jr.: Can we all just have some peace and quiet?

Shopkeeper: Hey, the reporter who wants to interview me before the shop transfer goes through is here. Would you like to take a picture together?

Shopkeeper: They're the ones who made this ice cream miracle happen—the young researchers from Laplace.

Thirty-seven: My hypothesis is correct. Medicine Pocket is lying. Chocolate ice cream isn't beans.

Thirty-seven: It's true that cocoa beans have "beans" in their name, but they're from the Malvaceae family, not the Fabaceae family. What's more, cocoa beans are processed into chocolate ice cream.

Thirty-seven: It's like getting a very bad negative infinite non-repeating decimal, like -1.414213562373 ...

Thirty-seven: It can't be written as a ratio of two integers, and the string of digits after the decimal point never ends. It looks awful.

Thirty-seven: However, if we square it—it makes 2! A favorable number!

Thirty-seven: So, there's nothing wrong with chocolate ice cream.

A Reply from Across the Waves: I hope you can try ice cream too. I asked Regulus to bring you some. Along with this letter, I've enclosed further papers, journal clippings, and printed materials I've collected. You can use these to update the island's database.

A Reply from Across the Waves: I've also gathered many new puzzles proposed by mathematicians from the outside. Some are extremely challenging, and I already have ideas for a few solutions.

A Reply from Across the Waves: But to be fair, I'll give you some time to work on them too. That way we can exchange our thoughts in the next letter.

A Reply from Across the Waves: And there's also this "newspaper." Mesmer found a report on me and suggested I send it so you can see how well I'm doing for yourself.

A Reply from Across the Waves: All in all, I'm happy here, and my research is progressing.

A Reply from Across the Waves: This time, I'll find a better answer. It might not be perfect, but I believe it will be good enough.

A Reply from Across the Waves: Then I'll return to the island, and Sophia will come back too. We'll all share the light of Truth, just like before.

A Reply from Across the Waves: You may have noticed that the total word count of this letter is a palindrome. I trust you understand what I mean; just don't try writing me a short reply. That would be cheating.

A Reply from Across the Waves: I look forward to your reply.

As the last stroke falls on the letter, the writer successfully completes her declared goal.

There is no need to count again. The reader confirms this to be the case.

Six: Hah.

Upon unfolding the letter, the young man unmistakably expresses his first sign of emotion. Of course, the other party present doesn't miss this significance of the moment.

Regulus: Woah! Did you just smile? I thought that pale, gloomy expression was welded to your face.

Sonetto: Regulus!

The One Who Counts Sand

DPL-08

The One Who Counts Sand

Sonetto: Sorry, Regulus didn't mean to offend. We meant to say ...

Six: 37 mentioned preparing a gift for me.

Regulus: Oh, right! You mean the ice cream.

Regulus: But you know, ice cream melts, right? I couldn't let my brand-new APPLe III get all sticky, could I?

Regulus: So I ate it for you. You can thank me if you want.

Six: Much obliged.

Regulus: Uh, well, you're welcome.

Now it's the ice cream thief's turn to feel awkward.

Sonetto: I'm sorry ...

Six: As for the other items 37 sent, please place them here. I'll put them away later.

Sonetto: Please remember to check them. I left the list here, too.

Regulus: Not to brag, but it was real hefty! Yeesh.

Regulus: We're in a hurry to get back to the Foundation, so hurry up and write a reply, and we'll take it back with us. Time ain't on our side.

Surprisingly, the young man shakes his head.

Six: There's no rush for now.

Regulus: You sure?

Regulus: Not to stick my beak in where it isn't welcome. That kid's a right looney at times, but she's still a kid at heart. Aren't you worried about her at all?

Six: 37 sent some puzzles. I need time to think through the answers.

It's an evasive answer, but the rockin' pirate simply stares into his expressionless face and, after a time, accepts it.

Regulus: Righto, if that's what you want.

Sonetto: Regulus?

Regulus: I'll go get the ship ready. My new partner and I need some time to bond. She has a bit of a touchy temper sometimes.

Sonetto: Understood. I'll wait here.

Sonetto watches her companion leave.

Sonetto: ...

Six: You seem troubled.

Sonetto: You're right.

The young man's gaze is as calm as still water. Under his silent scrutiny, Sonetto decides to voice her concern.

Sonetto: I've been thinking about 37. I'm not sure if keeping things the way they are is the right thing to do anymore.

Sonetto: She's brave. Even when facing the worst, she never backs down, leaving her home alone to find new answers in the outside world. I admire her.

Sonetto: I'm not here to judge your school's philosophy. Apeiron has provided us with numerous ways to combat the "Storm."

Sonetto: Only 37 still seems to believe that finding the Truth will solve everything, even what happened with Sophia and your fellow islanders.

Sonetto: But can that Truth truly solve everything?

Six: ...

The young man remains silent. Sonetto takes a deep breath and continues.

Sonetto: Maybe, sometimes, bad things just happen. People have to accept it and move on.

Sonetto: Someone needs to tell 37. Tell her those friends of hers who "returned to essence" won't come back. Tell her that maybe there is no Truth that can bring Sophia back.

Sonetto: While she's been searching for the Truth, we've avoided telling her it. Maybe that's not the best thing for 37. We're in danger of letting her waste her time and effort, only to find out it was all for nothing.

Sonetto: Maybe we should tell her before that happens.

Her composure buckles as she finishes each phrase. Sonetto knows she is crossing a line with these words. She looks up to observe the young man's expression.

6 shows no anger. He simply turns his expressionless face once more.

Six: There's no need. Everything you wish to say to her, 37 already knows.

Sonetto: You mean?

Six: You care about 37, and I'm deeply grateful for that, but there's a certain misunderstanding I must correct for you.

Six: She may still be a child, but she observes everything that happens in the phenomenal world with precision beyond any of us. 37 isn't blind. It's just that her faith remains strong, no matter what.

6 turns, facing out at the distant sea, but his gaze seems to pierce further and deeper beyond it.

Six: Allow me to put it this way: How many grains of sand do you estimate there are on this beach?

Sonetto: Sand?

The abrupt change of topic causes Sonetto's eyes to widen. She dutifully looks around the sprawling beach.

Sonetto: Perhaps a hundred million? A billion?

Sonetto: I'm sorry, I really don't know.

Six: I don't need an exact number, just an estimate within an acceptable range.

Six: There are numerous methods to obtain an answer—the Fermi estimation is one of them.

Sonetto blinks.

Sonetto: I've studied that ... Yes, you're correct.

Sonetto: Only ... sorry, I still don't understand what you're saying.

Six: Listen to my next question. How many grains of sand do you think there are on this planet?

The girl's eyes widen further.

Sonetto: The whole planet?

Sonetto: That ... must be a very large, astronomical number, something no one could possibly find—oh ...

For the first time, the young man nods.

Six: You see it now.

Six: This number is just "very large," but not "infinite." So it is possible to devise an approximate answer.

Six: Archimedes pointed this out over two thousand years ago. He claimed that even if sand filled the entire Cosmos, the number wouldn't exceed 10^{63}.

Six: His knowledge of the Universe was limited by his time, so while his estimation might not have been accurate, his method remains above reproach. The answer must be "many myriad," but not "infinite."

Six: It's finite.

Sonetto: So, then you mean ...

Sonetto looks at the sand beneath her.

The average mind cannot imagine numbers like 10^{63}, but this first-class graduate of the SPDM understands what the young man wishes to convey.

Sonetto: You believe that what 37 is searching for exists.

Sonetto: Yet, even if it exists in this world or anywhere beyond, finding it will be extraordinarily difficult.

Sonetto: But not impossible.

Sonetto lifts her eyes up from the not-quite-endless grains of sand, matching her gaze with that of 6's own.

Beyond the sea, a young girl is determined to calculate that very, very, very large number.

Six: I can confirm one thing.

Six: 37 made this hypothesis herself. Now, she is trying to prove it.

Six: And until she disproves it herself, no attempt at telling the Truth from any of us will shake her from it.